Scientometrics (2020) 123:113–142
https://doi.org/10.1007/s11192-020-03379-8

Scientometric analysis of social science and science disciplines in a developing nation: a case study of Pakistan in the last decade

Malik Muhammad Saad Missen[1] · Sajeeha Qureshi[1] · Nadeem Salamat[2] · Nadeem Akhtar[1] · Hina Asmat[1] · Mickaël Coustaty[3] · V. B. Surya Prasath[4,5,6,7]

Received: 13 June 2019 / Published online: 7 March 2020
© Akadémiai Kiadó, Budapest, Hungary 2020

Abstract
Our study is a scientometric analysis of research publications of science and social science disciplines in Pakistan during 2009–2018. The study examines 2000 published articles belonging to 50 research scholars of different disciplines. This analysis is conducted on three different levels: researcher level, field level and domain level. In this paper we discuss readability scores, title formats, single and multiple authorships of articles, citation rates, publication rates over time, the research contribution of both genders and the impact of PhD institution of authors in research publications. The IR features like number of words, sentence length and readability scores are extracted. The results indicate that more research is being conducted in science disciplines as compared to social sciences. It also appears that the readability scores of authors changed over time, and Science articles have higher readability scores compared to Social Science articles. Moreover, title formats are observed to effect citation rates. Titles tend to use more colons over time; and the use of simple sentences as titles is more than 50%, while use of question marks is below 3%, the rest are titles with colons. The citation rates also appear to increase due to co-authorship. Male researchers contribute more to research activities in science disciplines while in social science the ratio is almost equal while PhD institutes appear to influence the performance of researcher slightly. Finally, it is observed that the publication rates of Science disciplines are much higher than Social Science articles, depicting that the growth of Science disciplines is more rapid in Pakistan.

Keywords Scientometrics · Science · Social science · Decade · Country · Publication analysis

✉ Malik Muhammad Saad Missen
saad.missen@iub.edu.pk

Extended author information available on the last page of the article

Introduction

Academic writings in the form of books, research oriented journals and professional conferences have a long history. With globalization of the writings belonging to authors, malicious activities started. Therefore, during the 19th century research to identify academic writings begin. One of the earliest attempts was by Mendenhall (1887). This was followed by various researchers, who used word frequencies (Zipf 1932), CUSUM techniques (Morton and Michaelson 1990), and finally computer based methods for the solution of this problem. During 1990's research suggested text readability as a mechanism for author identification (Gillam 2013). Research community has focused on author profiling in the last decade due to availability of huge data online. Author profiling includes lexical features, characteristic features, syntactic features, and semantic features for identifying writing style of an author Stamatatos (2009). Various characteristics of the authorship profiling such as personality traits, age, gender, country of origin, native language, and degree of education can be ascertain by analyzing their writing styles (Mikros and Perifanos 2013). Feature extraction from texts to help in identifying characteristics of authors became the point of main focus in research. There have been many articles on this topic. Readability scores are currently being used to determine the author's age and gender (Weren et al. 2013; Schler et al. 2006; Weren et al. 2014; Patra et al. 2013; Rangel et al. 2013). Similarly, many other traits of author have been identified using various text features (Rangel Pardo et al. 2015). According to Gillam (2013) the longer words would be used with greater frequency at higher ages and female authors have a tendency to write in longer sentences than males. Practically author profiling is useful for criminal cases, marketing, in civil laws and much more. Owing to the importance of author identification, there is a need for much more research regarding feature extraction.

Due to various reasons, the effect on researchers writing style has been a debatable topic in the research community. Many researchers have studied the ways in which publications of author change with time. Effect of new technology in research writing is studied by various investigators (Hartley et al. 2001; Hartley 2007). There also have been studies on how title selection and use of colons effect citations (Gillam 2013; Buter and van Raan 2011; Nair and Gibbert 2016). It is observed that the use of colons and the length of titles vary with disciplines (Lewison and Hartley 2005; Subotic and Mukherjee 2014; Habibzadeh and Yadollahie 2010). The work of Glänzel and Thijs (2004), Franceschet and Costantini (2010) also supports the fact that co-authors in an article are a reason for more citations. Researchers also investigated how age causes a change in publication rates over time. Gillam (2013), Todorovsky (2014) support such studies for single author. Growth rates for science disciplines were also analyzed by researcher Tabatabaei-Malazy et al. (2016).

All the research conducted is documented and stored forming huge databases. Scientometric analysis is performed on such available documented knowledge to obtain quantitative results and determine the growth of knowledge in various fields. In Bornmann and Mutz (2015) such growth rates are observed. In Dias et al. (2018) a scientometric analysis is performed on 20M papers published in last three decades. The analysis studied the similarity between scientific fields on bases of linguistics, citation and expert classification. Similarly, Gupta and Mahesh (2013) examined the research work in social science in four Asian countries including Pakistan. The study analyzed 17,000 publications during 1996–2011. In Taşkın and Doğan (2013) an analysis on Turkish scholars is conducted. It examines 197,687 publications (during 1928–009) belonging to main scientific fields. The analysis studied half-life, multiple authorship, publication types and number of authors

in articles. Similarity between six reputed journals for five scientific disciplines was observed (Wolfram and Zhao 2014). Moreover, Paula et al. (2017) performs an analysis on social science in East Asia and Philippines. The analysis is based on publications of past eighteen years. The authors suggests that the older papers were more favored due to higher citation rates than recent papers.

This paper presents a comprehensive comparative analysis of research articles produced in the field of science and social science. However, only the articles published by research scholars working in academic institutes of Pakistan, during 2009–2018, are selected. Our analysis is conducted on 2000 published research articles. Our aim is to find variation in features like readability scores, citation rates, publication rates, effect of title format, and multi authorship over a period of 10 years. The effect of co-authors and title style on citations of papers is also analyzed in the paper. Readability scores are a marker to depict the change in an author's writing over time. Moreover, we have seen that feature extraction is the current topic of interest for author profiling. Hence, the selected research articles are also examined to extract the number of words, sentence length, passive sentences used, grade levels and readability scores. These information retrieval (IR) based features are extracted to provide a dataset that may be used by research scholars to determine characteristics of authors. The need to identify the authors of academic publications is essential; due to this fact the paper draws attention towards the author profiling by feature extraction also. A significant advantage of such features is that they can be applied to any language and any corpus with no additional requirements except the availability of a tool to segment text into tokens. Our study is unique in a way; as it performs a multi-aspect analysis of science and social sciences in hierarchical manner. We provide results by comparing articles at three different levels; researcher level, field level and at domain level. Hence, a clear picture of the results is obtained. This paper extends the work done previously by other researchers on writing over time. The paper includes eight objectives that are mentioned below:

- To extract readability scores of researchers and compare these readability features to record the improvements in writing over time.
- Investigate the effect on citations of articles due to title selection, use of colons and question marks in titles.
- Study the variation in use of colons and question marks in titles with passing years.
- Whether or not the articles written with co-authors cited more than those written by single author.
- Determine if the passing years were adding more to the achievements of researchers by analyzing publication rates.
- Compare the research contribution of both genders in Pakistan to conclude if there was any discrimination regarding gender.
- Observe if the researchers who did their PhD abroad were contributing to research more than those who did PhD in Pakistan.
- Study which domain (science or social science) is progressing rapidly in Pakistan.

Literature review

Our literature review covers the latest challenges in three domains of research. These domains are:

- Writing over time,
- Author profiling,
- Scientometric analysis.

Writing over time

Effect of new technology

The effect on researchers writing style due to various reasons has been a debatable topic in the research community. The ways in which publications of author change with time are studied by many researchers. The use of new methods and technology for writing articles has become common over time, which rose questions like how the computer aided technology has affected the writing styles of researchers. In Bornmann and Mutz (2015) the effect of new technology in research writing is studied. The author selected three authors for analysis the change in their writing styles over a period of 30 years. Each author provided a sample of their work and the details about how they changed their way of writing with time. The results indicated that there was no effect of use of new technology on writing styles of the three authors. The ways in which the three authors changed their tools for writing differ a lot but individual writings have no change over time.

Title formats and citations

The role of titles in an article is much essential as it affects the citation rates of articles. Many research articles examine use of colons and question marks in articles and also the affect of these on citation rates. The author in Todorovsky (2014) performed an analysis on the use of colons in titles. He selected articles of five years; belonging to cancer research across the world and UK science articles. The study also examined the variations in use of colons over time. According to him, the title length and colons in titles increased with passing years. However, other characteristics of titles have also been investigated by researchers; such as title length, number of words used, and punctuation usage (Stamatatos 2009). The analysis was based on 600 research articles of computer science taken from IEEE journals. He suggested that useful analysis can be conducted on titles by making raw lists of the word frequencies. The results show that some of the findings in literature about title format are correct but some concepts are misleading. Use of question marks in titles is also a question that has not yet been answered with certainty. In Stamatatos (2009) the author found only two articles with question marks from a dataset of 600 articles. The use of question marks is rare according to the analysis conducted on titles. It is observed that the use of colons and the length of titles vary with disciplines (Todorovsky 2014). The author selected the articles from various fields for analysis. Another study states that the title selection influences citation rates. In Weren et al. (2013) the author states that the use of non-alpha numeric characters has small effect on the citation rate of article. This study is based on a dataset that included articles from five top journals during 1997–2006. The article impact, citation count, author attributes and article attributes are examined in this article. Furthermore, Schler et al. (2006) clarifies that the studies on titles have generally been based on articles taken from journals, not on the articles of same field only. This analysis is based on psychology articles that were downloaded and examined for title characteristics. It used Scopus to get citations of each article and titles with colons were also analyzed but the author found that in this field the articles with more amusing titles were downloaded

more. Studies also suggest that articles with shorter titles are cited more (Schler et al. 2006; Weren et al. 2014). Similarly, Hartley and Cabanac (2015) inquired articles and book chapters of an author and concluded that the use of colons increased over time, citation rates were also dependent on title formats.

Single and multi-authorship

It is a common practice to write articles with co-authors, this may has affect citation rates according to various studies. Hartley and Cabanac (2015) analyzed publications of the past 50 years belonging to an author. They examined the citation rates of articles written by single author and those written with co-authors. The results indicated that the articles written with co-authors were cited more. Similar results were obtained by Patra et al. (2013) showing that the articles with co-authors were more likely to be cited. Both the studies are not diverse, rather limited to a specific dataset. More research could be done regarding this inquiry as citation rates are also affected by the year of publication and other characteristics. Furthermore, the work of Rangel et al. (2013) also supports the fact that co-authors in an article are a reason for more citations. In this study, different disciplines were examined and articles were gathered from Italian universities for analyzing collaboration impact on articles. The authors explained how collaboration impact changed for each discipline. Science disciplines had more collaboration than social science disciplines. However, computer science articles are in between the two domains.

Publication output over time

There has been research on the publication outputs obtained in specific fields in different geographical locations. These studies are useful and may help in policy making. The analysis of articles helps to find out how much progress has been made over time. Such studies help to point out the areas where more focus is required and points out the domains excelling in research. In Hartley and Cabanac (2015) Hartley studied how age causes a change in publication rates over time. His dataset includes the research articles and book chapters written by a single author during 50 years. Hartley observed that the publication rates of authors remain consistent only when their other activities are abandoned with age. Similarly, Gupta and Mahesh (2013) studies the overall contribution of one researcher in scientific research. Such researches were on single authors output. Another study (Rangel Pardo et al. 2015) suggests that scientific productivity increases with time. Tabatabaei-Malazy studied articles on herbal medicine published until 2015. The articles were gathered from Scopus. According to the results, the publication rate increase with time. This study is specific for just a single discipline and cannot be generalized. There was a need for generalized results on the productivity of research. Thus another contribution was by Bornmann and Mutz (2015), who analyzed all the disciplines of science to study the growth rates. Moreover, their work includes articles up to 2012, and used number of publications and their citation rates to depict growth (Dias et al. 2018). They observed that the growth rate in mid of eighteenth century was 1% which rose to 9% till 2012.

Author profiling

Author profiling also known as author attribution means to find characteristics and features of the text written by an author. The text is examined so that the data about author can be

gathered. This research area started in the nineteenth century with globalization of the writings belonging to authors (Mendenhall 1887). Research community has focused on author profiling in the last decade due to availability of huge data online. Author profiling includes lexical features, characteristic features, syntactic features, and semantic features for identifying writing style of an author (Taşkın and Doğan 2013). In Taşkın and Doğan (2013) the author presents a survey on author profiling, focusing on the methodologies being used and showing the progressive research in this domain.

The feature extraction from texts to help in identifying characteristics of authors became the point of main focus in research. There have been many articles on this topic. Readability scores are currently being used to determine the author's age. In Paula et al. (2017) the author profiling task was performed on blog to identify the age and gender of authors. The experiments included feature extraction, that could help to determine age and gender of authors. One of the features was readability scores, used for finding age of author. Younger authors are expected to use shorter words so these scores were helpful. The author achieved to classify 53% of English and Spanish authors only. Similarly, Gillam (2013) determined age and gender of ten thousand blogs. Multi-class real window algorithm was used for blogs. He used words, sentence length and content-based features for experiments; achieving 43.8% accuracy for age and 80.1% accuracy for gender. Extending his previous study, Weren et al. (2014) used 61 IR-based features in the experiments to identify which features were more useful for author identification. A corpus of 236 K blogs and readability scores was used to determine age of the writer. A similar study was conducted by Patra et al. (2013) wherein features like word length, word frequency, sentences, sentence length, punctuations, and pronouns to determine age and gender. An accuracy of 56.83% (gender) and 28.95% (age) was achieved. In Rangel et al. (2013) performed author profiling of twitter users. The use of content-based features is more promising for author identification. The author provides an overview of PAN 2013; it identifies age, gender and five personality traits of authors of four languages using various features.

Reviewing the research conducted on author profiling gives a clear image of how this field is progressing in the world. For practical use author profiling is useful for criminal cases, marketing, and in civil laws. Owing to the importance of author identification, there is a need for much more research regarding feature extraction. Our work includes a small dataset of 2000 articles of Pakistani researchers. These articles belong to 10 different fields and have been published during 2009–2018. These are examined to extract the words, sentence length, use of passive sentences and readability scores. These IR-based features are extracted to provide a dataset that may be used by research scholars to determine characteristics of authors. The need to identify the authors of academic publications is also essential; due to this fact our analysis draws attention towards the author profiling task of academic researchers.

Scientometrics

The field of Scientrometrics has gained much popularity in the past decades. Scientometric analysis is performed on available documented knowledge to obtain quantitative results. These studies determine the growth of knowledge in various fields. In Dias et al. (2018) a scientometric analysis is performed on 20M papers published in last three decades. The analysis studied the similarity between scientific fields on bases of linguistics, citation and expert classification. It suggests that the language has remained the same for disciplines while the individual linguistics has varied over time. The convergence

and divergence have remained in balance. Similarly, another work by Gupta and Mahesh (2013) examines the research work in social science in four Asian countries including Pakistan. The study analyzed 17,000 publications during 1996–2011. Scopus database has been used for gathering dataset and citations. The author gives research output of south Asian countries showing their growth rates, global ranking, publication rates and research priority. The highest global rank is obtained by Pakistan according to this analysis. These analysis have been conducted in different regions of the world, allowing researchers to reveal the productivity and other growth factors of disciplines. In Taşkın and Doğan (2013) an analysis on Turkish scholars is conducted. It examines 197,687 publications (during 1928–009) belonging to main scientific fields. The analysis studied half-life, multiple authorship, publication types and number of authors in articles. Wolfram and Zhao (2014) studies a similarity between six reputed journals for five scientific disciplines. Moreover, Paula et al. (2017) performs an analysis on social science in East Asia and Philippines. The analysis is based on publications of past 18 years. The author suggests that the older papers were more favored due higher citation rates than recent papers.

After studying all this literature we felt a need to combine some features from these three research areas and perform an analysis. Our study is a scientometric analysis of both science and social science disciplines that compares the growth of both domains in Pakistan. Such study has not been performed yet for the last decade and on basis of academic articles. Moreover, we have seen that feature extraction is essential and current topic of interest in author profiling but there has been no such study on science and social science articles of Pakistani scholars during past decade. Our study calculates the readabilities of articles for each author. These values can serve as a dataset for author profiling and may be refined more. Flesch readability score and Grade level tests have been identified as an essential feature for author identification. Our work provides these scores and compares them on three different levels; researcher level, discipline level and main domain level. Furthermore, studying the literature on writing over time we included more queries in our research extending the work done previously.

Methodology

Data collection

Domain selection

The main objective of this article is to give a scientometric comparison of the research work done in Pakistani universities in two major domains i.e. Sciences and Social Sciences. Therefore, these fields are chosen for collecting base data collection. These domains are further divided into 5 basic and the most researched fields. Selection of these fields is not a random selection but it is done after a detailed analysis of relevant faculties in Pakistani universities where these fields are found to be the most frequent in their relevant faculties. Selected fields are listed in Table 1.

Table 1 Science and social science disciplines selected for analysis

Science discipline	Social science discipline
Biology	Psychology
Chemistry	Economics
Physics	History
Mathematics	Education
Computer science	Political science

Author selection

In next step of data collection, we are supposed to choose researchers from each field. Since is not easy to decide which author's work to be chosen for data collection however still we did not decide to go for a random selection and defined a comprehensive criterion for this selection. For this purpose, we build an index of all researchers in every chosen field. The index contains all the required information about the researchers i.e. name, age, gender, year of PhD, specialization, institute of PhD, number of publications, etc. For relevant data collection, we have two options:

- To use any online indices (e.g., Scopus, Web of Science, CiteSeerX, etc) available for this purpose,
- To explore the web sites of educational institutes, profile pages of the researchers or their personal web sites, or any related web page for the author that could be useful for finding relevant information.

While the second option is difficult but we go with this one because it provides more precision in its nature. Online indices often do not track every researchers data or miss many of their publications. All the university websites of Pakistan were searched to find the required researchers. Once we have completed the designed index with relevant information, we filter this index with two filters i.e. (1) removing all those doing PhD after year 2008, (2) removing all those who are not currently serving in any public or private educational or research institute of Pakistan.

The rest of the data is sorted using the total number of articles published for each field. We select total 5 researchers from each field considering the following parameters:

- Male and female researchers are chosen in ratio of 3:2. This ratio is set because there are less women researchers with considerable work available than male researchers. However, equal number of male and female researchers are considered when comparing work of male and female researchers in this article to avoid bias.
- Equal number of researchers having PhD from Pakistan and those having from abroad.
- Articles written only in English are considered for research.

We then collected the research articles of each researcher that were published during period (2009–2018), see Fig. 1. Several Internet resources like Google scholar, ResearchGate (for obtaining free full texts), etc are used for downloading these articles. Regarding the difference in the number of articles of each researcher, we have to fix a range to avoid biases in our analysis. The maximum range of publications per researcher is fixed as 60 articles i.e. almost 6 articles per year. If in a year the articles were less than 6 then this

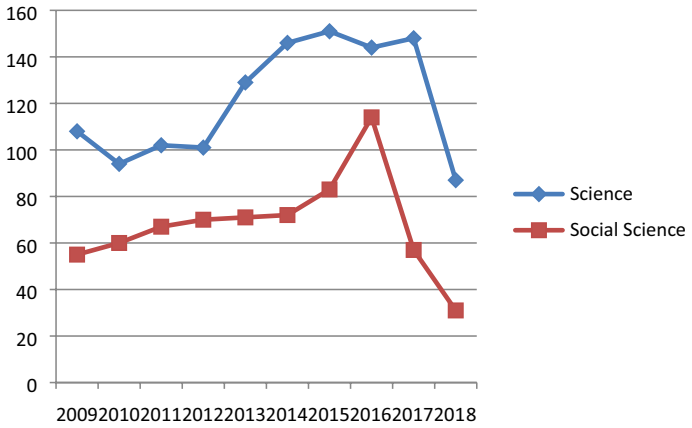

Fig. 1 Science and social science articles gathered from 2009–2018

number is completed by taking publications from the preceding year. This threshold of 60 articles is set according to the availability of articles on the internet. The minimum number of articles is set differently for science and social science due to difference in availability of publications. For science disciplines the minimum number of articles is set as 15 and for social science it is set as 10. We have to fix these maximum and minimum limits to maintain coherency between number of articles because of less number of articles in social science discipline and more number of articles in science discipline. The latest articles of a researcher for a particular year are selected for this research. In Pakistan, there is more research currently being done on science disciplines than in social sciences. Hence, there was a large pool of available researchers for science fields. The majority included male researchers while females with PhD degree and more than 15 publications were fewer. A total of 1210 research articles of science disciplines (almost 49 articles per researcher) and 680 research articles of social science disciplines (almost 28 articles per researcher) are downloaded for analysis.

Results and discussions

Readability scores

Research articles are a source to spread knowledge globally. Therefore, it is essential that these articles must comprehensive and easily understandable. This understandability helps in measuring Flesch-Readability scores (Flesch 1948). The Flesch-Readability score generates values between 0 and 100. The articles with higher scores are considered easier to read and those with lower values are difficult to read. These scores are computed using total number of words, sentence length, number of sentences, and passive sentences present in a document.

This feature can be used for any language which makes it more useful for identifying authors. We performed our analysis on three different levels; researcher level, field level and main domain level. At each level statistics were recorded and results were obtained. It

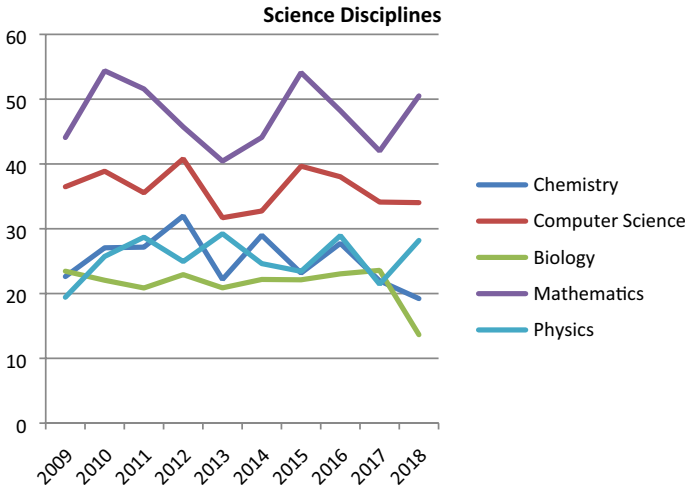

Fig. 2 Comparison of Flesch-readability scores of science disciplines

is to be noted that full text of the research articles is used to calculate the readability scores of the articles. The articles available in PDF form are converted to text form by using an online tool.[1] After conversion of the articles to text form, we perform manual auditing/cleansing of the full text. We remove the references, tables, figures, and mathematical expressions from the text and rest of the text is used for calculation of readability scores. Microsoft (MS) Word is used for calculation of readability scores. MS Word provides total number of words used, total sentences, percentage of passive sentences, Flesch Reading Ease, and the Flesch–Kincaid Grade Level of the article's text.

First we performed this study on 1206 articles, written by 25 researchers, belonging to five fields of science. The results depict how the articles of each researcher varied with time in terms of ease of readability. This data shows quite a difference in the authors of different fields while slight differences in an author's own writing over time. The use of more passive sentences makes articles tough and less likely to be understood by majority of readers. We then, performed the analysis on field level. The fields of each domain are studied to see the changes in text readabilities of their articles, during last decade and inquire which fields show more improvements. Figure 2 shows the Flesch scores for five fields of science domain. The results depict that articles of Mathematics are easiest to read and understand as their Flesch scores are above 40. Then we observe the curves of Computer Science articles in the graph; having scores in range 32–42. The articles of Physics and Chemistry show overlapping scores in the range of 20–30. These articles are difficult to read as their readability is below 30. Biology articles have the lowest scores in range of 20–25. These articles remained nearly same except for 2018, where the Flesch scores lowered to 12. In the graph, Mathematics line has mountain curves repeatedly; from 2009 to 2013 the scores improved and then again the values lowered. The same pattern is observed from 2013 to 2017. For Physics and Chemistry the scores remain in the same range varying a little every year.

[1] http://www.zamzar.com

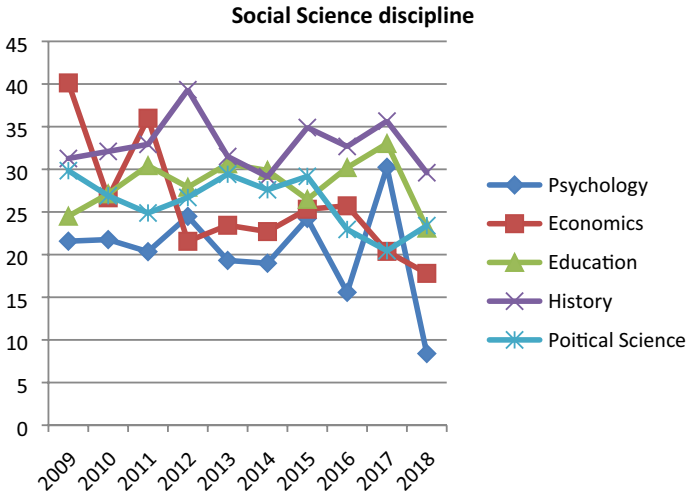

Fig. 3 Comparison of Flesch-readability scores of social science disciplines

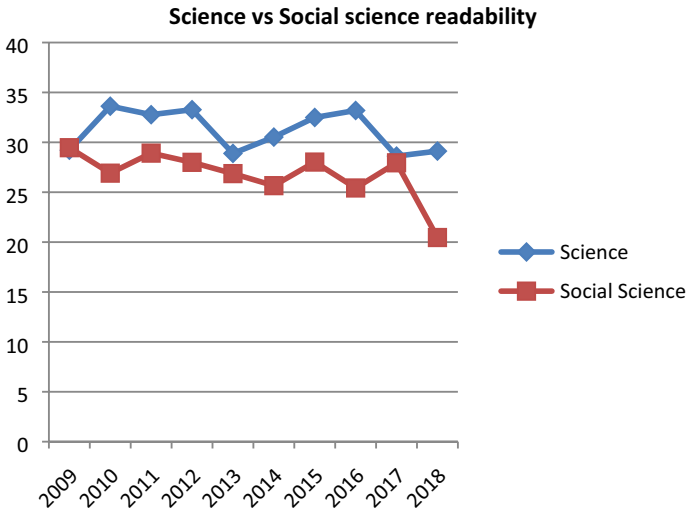

Fig. 4 Comparison of Flesch-readability scores of science and social science disciplines

Similarly, we compared the five fields of social sciences for their readability scores, see Fig. 3. It is clear that the readability scores of history and above 30. Whereas, the articles of Psychology have lowest scores but in the last 2 years there is improvement in the text readability scores. On the other hand, Economics articles are becoming difficult to read over time as their readability scores are lowering every year. The articles of Education do not show considerable improvement except for the year 2017. Over all its articles are below 30 which means these are not easy to understand by everyone. The last field under observation is Political Science. The articles of this field have become tough to read over time.

The third level involves comparing both the domains; Science and Social science. It is obvious from our results in Fig. 4 that the articles of science have higher readability scores in comparison to articles of social science in the given data collection. While there are minor variations in readability scores over time for both disciplines, overall science articles score remains between 30 and 35 while score for social science articles floats between 25 and 30. It can be observed that readability scores for science articles are generally improving over time except for years 2013 and 2017 while there is a likeliness of reduction in readability scores for social science articles over the passage of time. If we interpret these results in perspectives of Flesch score interpretations then science articles are classified as "Difficult to read" while social science articles can be classified as "Very difficult to read". Hence, in relative terms for the given data collection, it can be said that science articles are slightly better than social science articles in readability.

The results of our analysis depict that in Pakistan the articles of science disciplines are easy to read compared to the articles of social science disciplines. Changes in the text readability of articles suggest that the researcher's performance and writing skills are affected over time.

Change in article title over time

About 510 science articles and 440 social science articles from our dataset were analyzed to see the three main types of titles used by researchers in there publications. We divided titles into basic types; simple sentences, titles with colons, and titles with question marks. Articles having title of each type were separated. The results obtained are shown in Table 2. In science 61% titles were simple sentences, 38% titles with colons, and only 0.6% titles having question marks. While in social science there were 56% titles with simple statement, 41% titles with colon signs, and 3% titles with question marks. It is clear from our result that in both fields use of question marks is minimal. We observed, after comparing both domains, that there is more use of special characters (i.e. colons and question marks) in social sciences while science articles have mostly simple titles.

Our dataset was again analyzed to find out how time affected the type of titles used in articles. Therefore we divided years into five groups. For every group the titles with colons and question marks were recorded. This survey was also performed on both domains; sciences and social sciences. Aim of this study is to find out how the use of these special characters has changed over time. The obtained results are shown below in Table 3. For science domain, we observed that the use of titles with colon is increasing over time. The use of question marks is negligible. Furthermore, in social sciences the use of titles with colons is rapidly increasing over time. There are more articles with question marks in social science as compared to science fields.

Table 2 Articles having different titles types, belonging to science and social science 2009–2018

Domains	Simple	Colons	Question marks
Science	313	194	3
Social science	246	181	13

Table 3 Colons and question marks used in titles of articles over time 2009–2018

Years	Colons	Question marks	Colons	Question marks
2009–2010	24	1	27	3
2011–2012	24	1	33	2
2013–2014	31	0	41	5
2015–2016	39	0	56	3
2017–2018	27	1	29	0

Table 4 Citation rates for three types of titles of science articles

	Simple sentence	Colons	Question marks
Median (M)	22	32	6
Citation range	0–248	0–301	3–39
Number of articles (N)	313	194	3

Table 5 Citation rates for three types of titles of social science articles

	Simple sentence	Colons	Question marks
Median (M)	2	1	1
Citation range	0–206	0–93	0–54
Number of articles (N)	246	181	13

Title formats and citation rates

The next study is on the title type and its effect on citation rates of articles. The same dataset, used in previous study, is analyzed to find out the relationship between citation rates of articles and type of title chosen. In Table 4 the medians, and range of citations for types of titles in science articles are calculated. The median for titles with colon is 32. This is higher than other two types of titles. Field of biology has more use of colons in its titles. Citation ranges for simple titles and those with colon are very wide except for titles with question marks. Another observation is the use of more simple sentences in titles. Hence, the results show that there is more use of simple sentences in science articles but on field level it is observed that biology and physics use more titles with colon. The rate of citations for titles having question marks is lower showing that these are less preferred type of titles.

In Table 5, results obtained for social science disciplines are presented. Here we have medians of citation rates for all 3 types to be nearly equal. The citation range is

wide for simple sentence titles while the other two types have citations below 100. We also observed the difference in number of titles with each type; simple sentences in titles are mostly used and have high citation rates. Use of titles with question marks is reduced. In social sciences Economics has the use of maximum titles with colons and question marks.

Single and multi-authored citation rates

Using our data collections, we were able to examine another question under debate in literature i.e., how the citations of articles written by single author and those written with colleagues differ? We separated all the articles into these two sets i.e. single authored articles and multi-authored articles. After this the citation rates for every article were taken from Google Scholar. We compared the data for each field of science and social science. The medians, citation range, and number of articles examined for each discipline. The results indicate that the citation rates are influenced due to multi-authorship. However, the citation rates are also dependent on the date of publication i.e. the older the paper the more cited it is. Keeping this point in mind, we performed another round of analysis on articles by taking top cited four papers from each discipline (two with single author and two having multiple authors). The results are concluded in the Tables 6, 7, 8, 9 which show that the citation rates of articles having multiple authors are higher. Specific details of the articles with these two sub-groups single author, and multiple author are given in "Appendix". Tables also include a column for Normalized Citation Rate (NCR) which is computed by dividing the number of total citations by age of the article i.e., number of years till year 2019. NCR is used to know if older articles get more citations than the newer ones.

Publication rates over time

The publication rates of both domains; Science and Social sciences are discussed in this section. This study is aimed to find if there is an increase in the publications of authors over time. Studies show that publication rates of authors decline over time as the author becomes older (Weren et al. 2013). We analyze yearly published articles of authors of both domains mentioned above. The results are plotted in graphs, Figs. 5, 6, 7, 8, 9, which show the yearly progress of each author. All five disciplines of science are compared on basis of published articles per year. The publications of mathematics are highest, followed by biology publication rates. Furthermore, the publication rates of physics and chemistry are overlapping in the graph. Therefore we can say these two fields are similar in publication rates. Computer science publications per year are minimum and not increasing over time.

It is clear from the results plotted in Fig. 7 that, the publications of Economics' articles are highest. The publication rates are increasing per year except for the last two years; where we observed a clear decline in the publication rates. However, publication rates of Psychology and Political science increased over time. We observe that the published articles of Education increased in first five years but, then started declining. History shows minimal changes in its publications over time. From the above graph, we deduce that the

Table 6 Citation rates of science articles by single author

Citation	Year of publication	Normalized citation rate (NCR)	Titles of article
156	2009	15.6	Extended general variational inequalities
63	2009	6.30	Projection methods for nonconvex variational inequalities
33	2011	4.12	On a generalization of uniformly convex and related functions
9	2011	1.12	Prediction of GPCRs with pseudo amino acid composition: employing composite features and grey incidence degree based classification
3	2011	0.38	Living with violence in the home: exposure and experiences among married women, residing in urban Karachi, Pakistan
2	2014	0.40	Y 3 Fe 5 O 12 nanoparticulate garnet ferrites: comprehensive study on the synthesis and characterization fabricated by various routes
1	2015	0.25	Ethnobotanical analysis on the Campus of Govt. Degree College Rajouri (J&K), India
1	2009	0.10	The influence of computer assisted instruction (CAI) in chemistry on the Students' Achievement at higher secondary level in Bahawalpur
0	2016	0	Construction of mammalian expression vector of core gene of HCV of Pakistani isolate genotype 1a
0	2013	0	Genetic predisposition of NKX2. 5, GATA4 and CFC1 genes for selected congenital heart diseases in Pakistani population
0	2007	0	Active scheduling for programmable routers in open adaptive network environments
0	2010	0	Reliable data management framework for flash storage systems

Table 7 Citation rates of science articles by multiple authors

Citation	Year of publication	Normalized citation rate (NCR)	Titles of article
302	2011	37.75	Polyamines: natural and engineered abiotic and biotic stress tolerance in plants
182	2009	18.2	A survey of service discovery protocols in multihop mobile ad hoc networks
174	2012	24.85	Thermodynamics in f (R, T) theory of gravity
145	2014	29.00	A recent survey of reversible watermarking techniques
144	2009	14.4	Some relatively new techniques for nonlinear problems
131	2010	14.5	Hepatitis C treatment: current and future perspectives
125	2011	15.62	Intimate partner violence in urban Pakistan: prevalence, frequency, and risk factors
102	2011	12.75	Traditional uses of medicinal plants of Nandiar Khuwarr catchment (District Battagram), Pakistan
88	2010	9.77	Structural, electrical and dielectric properties of yttrium substituted nickel ferrites
83	2010	9.22	Aflatoxin M1 contamination in milk from five dairy species in Pakistan
72	2011	9.00	Effect of dipolar and exchange interactions on magnetic blocking of maghemite nanoparticles
65	2015	16.25	Traditional uses, phytochemistry, and pharmacology of Olea europaea (olive)

Table 8 Citation rates of social science articles by single authors

Citation	Year of publication	Normalized citation rate (NCR)	Titles of article
75	2010	8.33	Perceived stress, academic workloads and use of coping strategies by university students
72	2011	9.00	The contribution of individual variables: job satisfaction and job turnover
37	2010	4.11	Corruption and its deep impact on good governance in Pakistan
12	2012	1.71	On fuzzy order relations
3	2012	0.50	Facet-specific job satisfaction and experience of elementary school head teachers in Pakistan
3	2011	0.37	Wavell and the dying days of the Raj: Britain's Penultimate Viceroy in India
3	2010	0.33	Empire, law and history: The British origin of modern historiography of South Asia
2	2013	0.33	Silencing the past: role of the army in 1970–71 conflict and its depiction in Pakistani textbooks
1	2013	0.16	The Taliban Imbroglio: search for a durable peace in Pakistan
0	2015	0.00	Impact of positive psychological capital On work attitudes, intentions and behaviours among university teachers
0	2016	0.00	An analysis of the impact of Government size on economic growth of Pakistan: an endogenous growth
75	2010	8.33	Perceived stress, academic workloads and use of coping strategies by university students

Table 9 Citation rates of social science articles by multiple authors

Citation	Year of publication	Normalized citation rates (NCR)	Titles of article
206	2010	22.88	Working capital management and corporate performance of manufacturing sector in Pakistan
170	2013	28.33	TOPSIS for hesitant fuzzy linguistic term sets
88	2009	8.80	Psycho education and the family burden in schizophrenia: a randomized controlled trial
48	2010	5.33	Parental involvement in children academic motivation
46	2010	5.11	Job characteristics as predictors of work motivation and job satisfaction of bank employees
14	2012	2.00	Balochistan (from strategic significance to US involvement)
10	2014	2.00	Mindful eating and it's relationship with mental well-being
2	2010	0.22	Coalescing the romance of antiquity, literature, orient and imperial justice: Sir William Jones and the Birth of 'Indology'
1	2015	0.25	Islamization in Pakistan: an overview
0	2018	0.00	One unit scheme: the role of opposition focusing on Khyber Pakhtunkhwa
0	2013		A factor in East Pakistan's separation: political parties or leadership by Rizwan Ullah Kokab, Massarrat Abid appraisal of the change in spatio-temporal

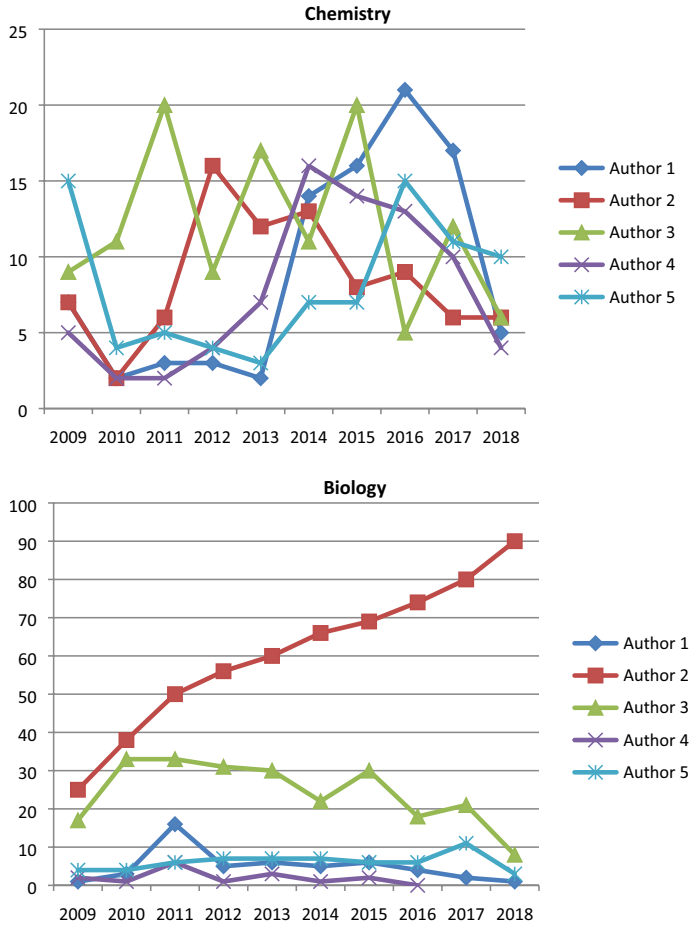

Fig. 5 Chemistry and biology published articles of researchers during 2009–2018

publication rates of every field have declined in the last two years. Concluding our analysis, we represent growth rate of both domains in line graph, see Fig. 10. There is a clear difference in the two lines of graphs showing publications of science and social sciences. The publication rates of science are very high compared to those of social sciences, in Pakistan. There is an increment in the articles published each year over time.

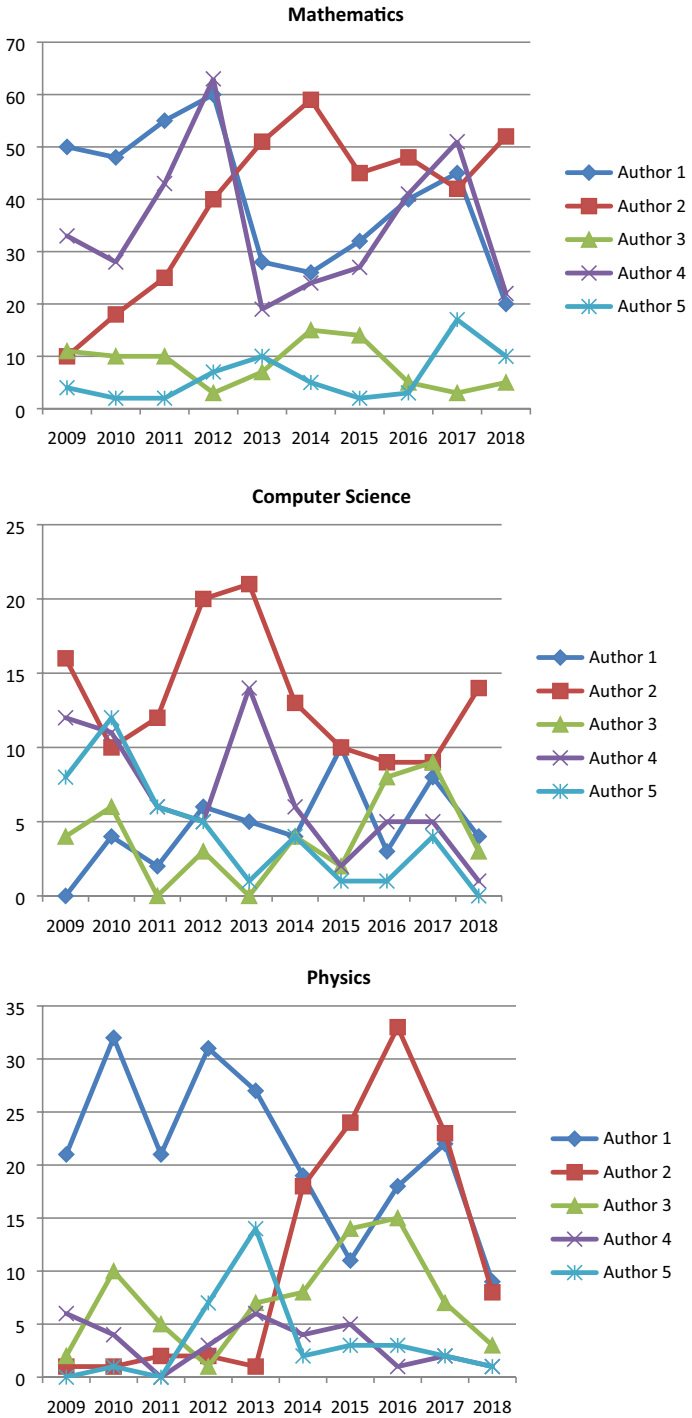

Fig. 6 Mathematics, computer science and physics published articles of researchers during 2009–2018

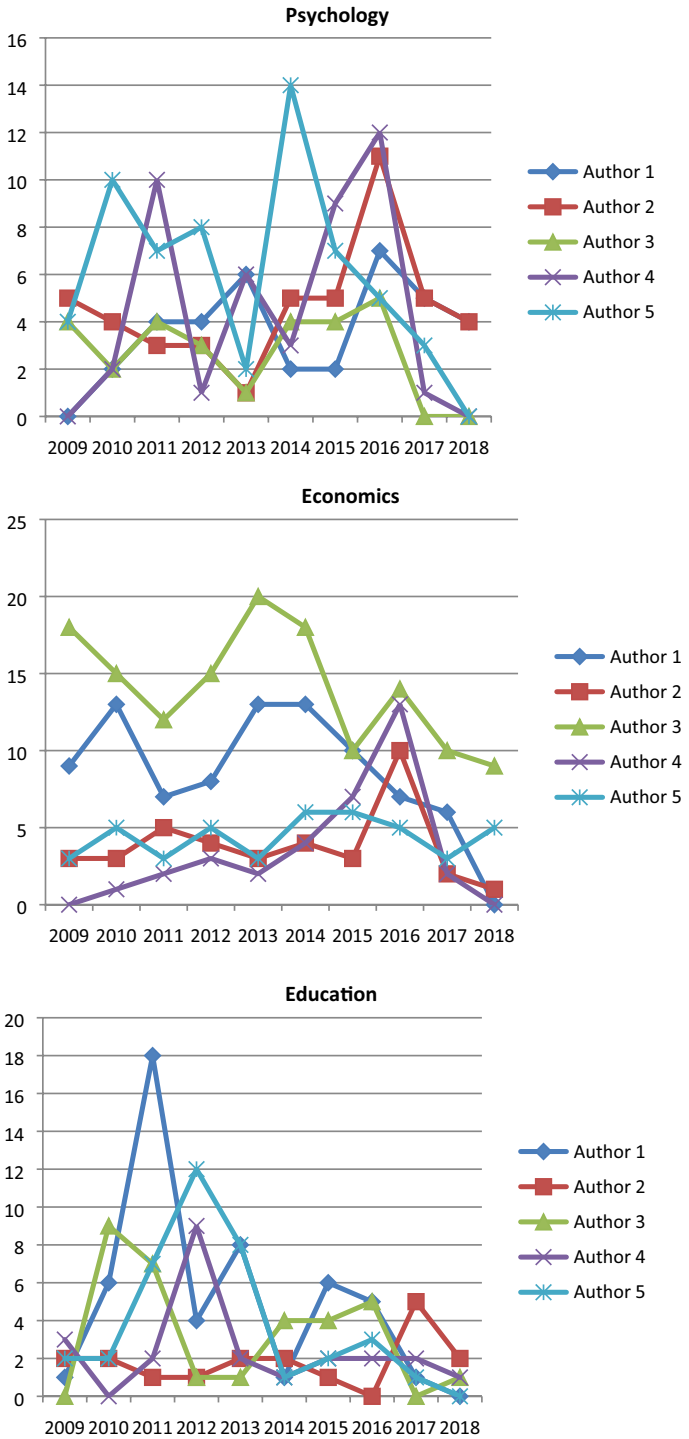

Fig. 7 Psychology, economics and education published articles of researchers during 2009–2018

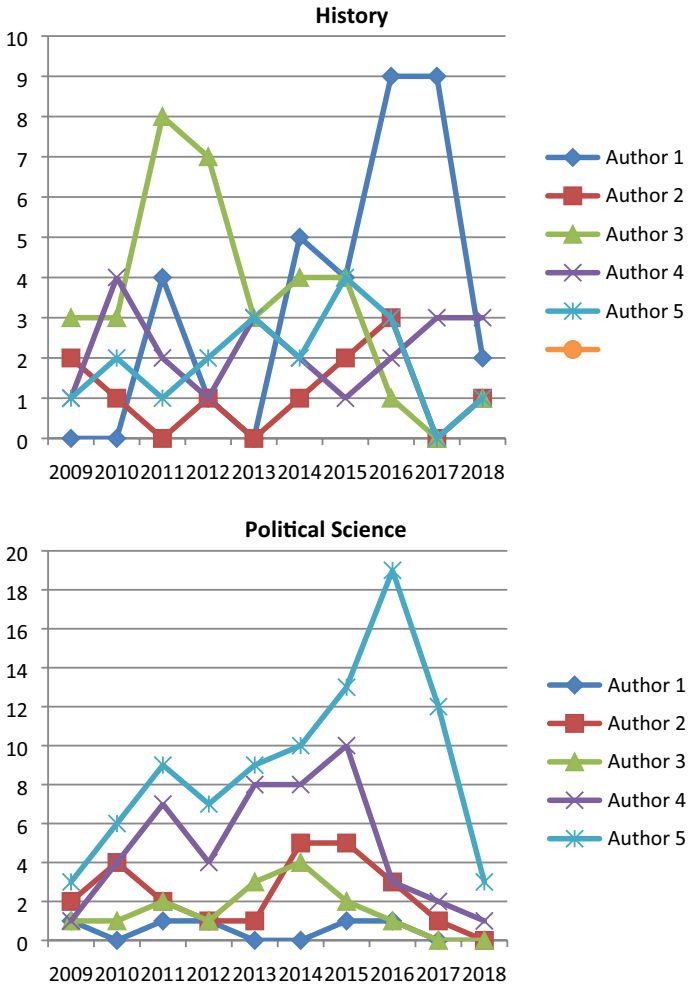

Fig. 8 History and political science published articles of researchers during 2009–2018

Contribution of both genders in research publications

This study examines the contributions of both genders in the academic research in Pakistan. For our analysis we selected two male researchers and two female researchers from our dataset. The comparison of publication rates of both genders was performed and results are shown in Fig. 11 below. The graph shows the research articles published per year in science disciplines by both male and female researchers. It is obvious from the graph that articles by male researchers are published more than female researchers. The

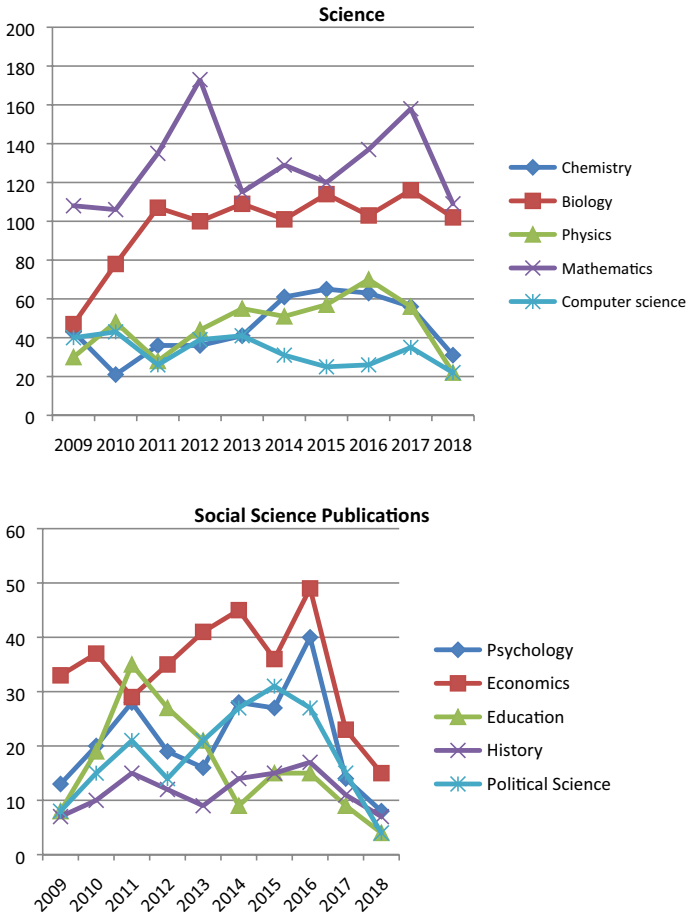

Fig. 9 Comparison of science and social science disciplines published articles of researchers during 2009–2018

publication rates for male researchers are increasing over time. There have been various researches on this difference of contribution in research by both genders. The reasons for fewer publications by female researchers may be that there are much more responsibilities for females as they get married, or receiving less funding, or focusing more on quality of work than the quantity; so they cannot contribute much in the research (Larivière et al. 2011).

Similarly, the articles for both genders were compared in fields of social sciences. The results are shown in Fig. 11. It is observed that the articles of women are greater than men in the fields of social science. Hence, we can say that women researchers are less active than male researches in science fields while in social sciences females are contributing more to the research.

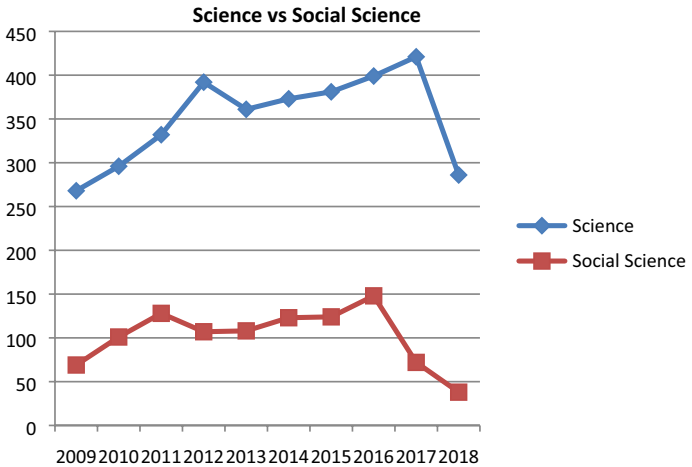

Fig. 10 Comparison of science and social science disciplines published articles of researchers during 2009–2018

Effect of PhD institute on performance of researcher

The articles of scholars are examined on the bases of their PhD institutions. The question that whether getting PhD degree from Pakistan or from a foreign country has effect on the publication rates of researcher. For this analysis, we divided our data into two categories researchers from each discipline were taken in both categories according to their PhD institutes. The results of last 10 year articles are shown for science and also social sciences in the form of graphs in Fig. 12. The results for science disciplines show that PhD from foreign country increases the productivity rate. However, the scholars of social sciences who studied in Pakistan show more publications per year. But these results cannot provide all the dimensions related to publication rates. Other factors like discipline choice, quality of work, personal abilities, current post can also effect rate of publication. This analysis can further be refined by taking in account other activities of author and increasing the data set used.

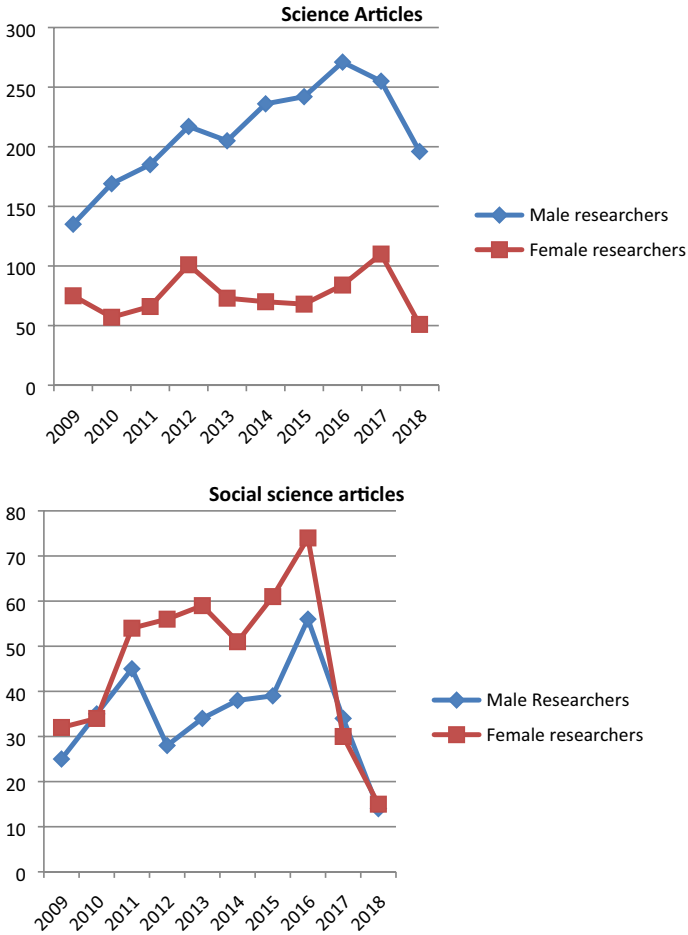

Fig. 11 Comparison of science and social science results for publication rates of male and female researchers

Limitations of the study

While the current study gives detailed analysis of the data collection used in perspectives of several research questions, it still suffers from a few limitations that are listed here:

- This study does not fully make use of any bibliographic databases like Web of Science, Scopus etc. While this was more a requirement of the study than a limitation, however we will try to tackle this issue in our future work.
- While this study covers enough scope by focusing on two major disciplines and related sub-disciplines, it somehow falls short on number of articles to be analyzed. We plan to increase the number of articles by incorporating further sub-disciplines.

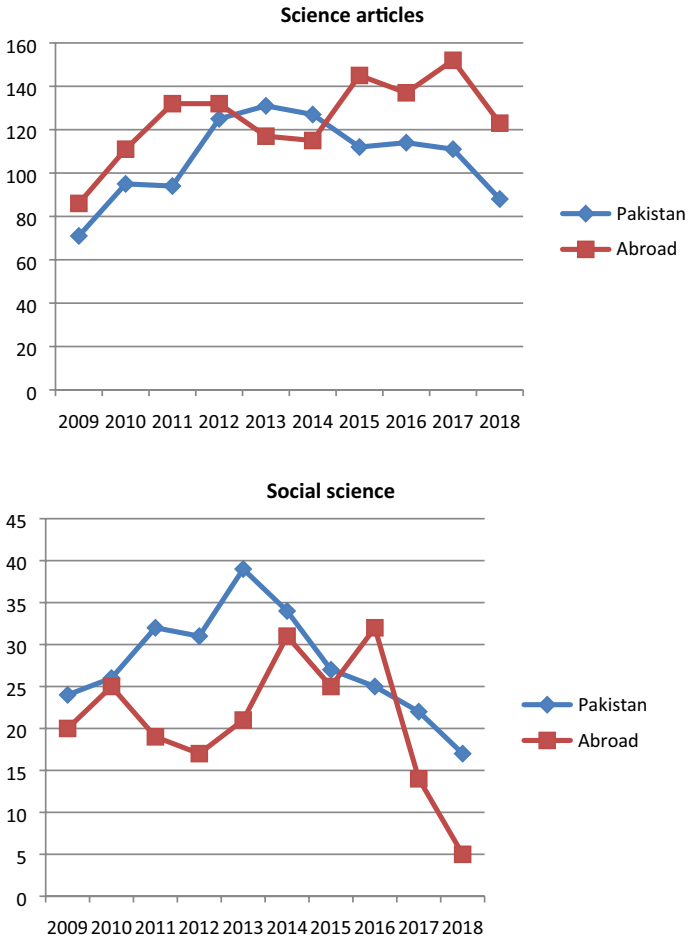

Fig. 12 Comparison of science and social science publications by researchers who did PhD in Pakistan and abroad

Conclusions

Scientometric analysis of research articles produces helpful results that prove to be useful in policy making and author profiling task. The results of our inquiries suggest that Science disciplines are progressing more in Pakistan. The research activities in social science are also increasing over time but the progress is slow and comparing it with science we might say that social science needs to flourish more in this country. The readability scores of science are higher than social science. The articles of mathematics and computer science have high scores and easier to read. The readability of these articles is not consistent over

time. However, in social science only history shows scores above 30, all other disciplines have low average scores. Other readability statistics, such as words and sentences of each article are also given. These features may help in author identification in future. Regarded title formats of articles we examined that in science 61% titles were simple sentences, 38% titles with colons, and only 0.6% titles having question marks. While in social science there were 56% titles with simple statement, 41% titles with colon signs, and 3% titles with question marks. The use of colons and question marks has increased over time in social science articles. Citation rates of colons are slightly higher while titles with question marks are not cited much. Science titles are mostly simple statements except for biology where colons are used frequently. Moreover, the articles written with co-authors are cited more than articles written by single author. This is true for both science and social science domain. An increase is observed in multi-authorship over the last decade.

The results of analyses of the growth rate of both domains; Science and Social science depict that in Pakistan science disciplines have publication rate very high compared to social science. Furthermore, the disciplines of Mathematics and economics show highest output over the last decade. There is a need for making such policy that will promote research activities in Pakistan. Overall, the publication rates of both domains are rising over time. After examining the research output of male and female research scholars we conclude that in science disciplines female researchers are less active, giving comparatively low output than male researchers. However, in social science disciplines especially psychology, female output is higher than male researchers. The reason for this may be less interest of females in technical fields like computer science, physics etc. Female researchers seem more active in research in theoretical fields. In science disciplines, the researcher having PhD degree from a foreign institute show more output per year generally while, in case of social sciences the results are inverse. But this inquiry still needs more examining on a larger dataset and including other factors too that may influence the publication rates.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Appendix: Medians, citation range, and number of articles

For the analysis conducted in this work, we had to consider every article written by the selected authors in the past 10 years. A total of 2351 articles of science disciplines and 1176 articles of social sciences were considered. These were separated in two groups; those with single authorship and multi-authorship. The medians, citation range, and number of articles examined are given in the Table 10. Note that these results for each field show that there are very few articles written by single author, except in Mathematics where there are considerable number of single author articles relative to other fields, however they too are less than multiple author based articles.

Table 10 The medians and ranges of citation rates for single and multi-authored articles in different fields

	Single author	Multiple author
Physics		
Median citation	1	22
Citation range	0–2	0–86
Number of articles	5	335
Biology		
Median citation	8	21
Citation range	0–15	0–646
Number of articles	17	550
Chemistry		
Median citation	1	22
Citation range	0–2	0–86
Number of articles	5	335
Mathematics		
Median citation	42	140
Citation range	0–189	0–416
Number of articles	150	600
Computer science		
Median citation	5	32
Citation range	0–60	0–182
Number of articles	11	343
Psychology		
Median citation	0	1
Citation range	0–75	0–103
Number of articles	13	127
Economics		
Median citation	3	2
Citation range	0–94	0–106
Number of articles	49	89
Education		
Median citation	2	22
Citation range	0–76	0–302
Number of articles	8	210
History		
Median citation	1	22
Citation range	0–2	0–86
Number of articles	5	335
Political science		
Median citation	1	22
Citation range	0–2	0–86
Number of articles	5	335

References

Bornmann, L., & Mutz, R. (2015). Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references. *Journal of the Association for Information Science and Technology*, *66*(11), 2215–2222.

Buter, R. K., & van Raan, A. F. (2011). Non-alphanumeric characters in titles of scientific publications: An analysis of their occurrence and correlation with citation impact. *Journal of Informetrics*, *5*(4), 608–617.

Dias, L., Gerlach, M., Scharloth, J., & Altmann, E. G. (2018). Using text analysis to quantify the similarity and evolution of scientific disciplines. *Royal Society Open Science*, *5*(1), 171545.

Flesch, R. (1948). A new readability yardstick. *Journal of Applied Psychology*, *32*(3), 221.

Franceschet, M., & Costantini, A. (2010). The effect of scholar collaboration on impact and quality of academic papers. *Journal of Informetrics*, *4*(4), 540–553.

Gillam, L. (2013). *Readability for author profiling? Notebook for PAN at CLEF 2013*. Working notes for CLEF 2013 conference, Valencia, Spain. Available online at http://ceur-ws.org/Vol-1179/CLEF2013wn-PAN-Gillam2013b.pdf.

Glänzel, W., & Thijs, B. (2004). Does co-authorship inflate the share of self-citations? *Scientometrics*, *61*(3), 395–404.

Gupta, B. M., & Mahesh, G. (2013). A comparative analysis of social sciences research publications in four South Asian countries. *Library Philosophy and Practice* (e-journal). 956. https://digitalcommons.unl.edu/libphilprac/956.

Habibzadeh, F., & Yadollahie, M. (2010). Are shorter article titles more attractive for citations? Cross-sectional study of 22 scientific journals. *Croatian Medical Journal*, *51*(2), 165–170.

Hartley, J. (2007). Longitudinal studies of the effects of new technologies on writing: Two case studies. *Studies in Writing*, *20*, 293.

Hartley, J., & Cabanac, G. (2015). An academic odyssey: Writing over time. *Scientometrics*, *103*(3), 1073–1082.

Hartley, J., Howe, M., & McKeachie, W. (2001). Writing through time: Longitudinal studies of the effects of new technology on writing. *British Journal of Educational Technology*, *32*(2), 141–151.

Larivière, V., Vignola-Gagné, E., Villeneuve, C., Gélinas, P., & Gingras, Y. (2011). Sex differences in research funding, productivity and impact: An analysis of Québec university professors. *Scientometrics*, *87*(3), 483–498.

Lewison, G., & Hartley, J. (2005). What's in a title? Numbers of words and the presence of colons. *Scientometrics*, *63*(2), 341–356.

Mendenhall, T. C. (1887). The characteristic curves of composition. *Science*, *9*(214), 237–249.

Mikros, G. K., & Perifanos, K. (2013). Authorship attribution in greek tweets using author's multilevel N-gram profiles. In *AAAI spring Symposium: Analyzing Microtext*.

Morton, A. Q., & Michaelson, S. (1990). *The QSUM plot* (Vol. 3). Edinburgh: University of Edinburgh, Department of Computer Science.

Nair, L. B., & Gibbert, M. (2016). What makes a 'good' title and (how) does it matter for citations? A review and general model of article title attributes in management science. *Scientometrics*, *107*(3), 1331–1359.

Patra, B. G., Banerjee, S., Das, D., Saikh, T., & Bandyopadhyay, S. (2013). Automatic author profiling based on linguistic and stylistic features. In *Notebook for PAN at CLEF*.

Paula, Antoinette G. Acuña, Alfonso, Carlo J. Ellasos, Jethro, A. Bautista, & Madhavi, Devaraj. (2017). Scientometric mapping of social science studies in East Asia and the Philippines. *International Journal of u- and e- Service, Science and Technology*, *10*, 49–58. https://doi.org/10.14257/ijunesst.2017.10.8.05.

Rangel, F., Rosso, P., Koppel, M., Stamatatos, E., & Inches, G. (2013). Overview of the author profiling task at PAN 2013. In *CLEF conference on multilingual and multimodal information access evaluation* (pp. 352–365). CELCT.

Rangel Pardo, F. M., Celli, F., Rosso, P., Potthast, M., Stein, B., & Daelemans, W. (2015). Overview of the 3rd author profiling task at PAN 2015. In *CLEF 2015 evaluation labs and workshop working notes papers* (pp. 1–8).

Schler, J., Koppel, M., Argamon, S., & Pennebaker, J. W. (2006). Effects of age and gender on blogging. In *AAAI spring symposium: Computational approaches to analyzing weblogs* (Vol. 6, pp. 199–205).

Stamatatos, E. (2009). A survey of modern authorship attribution methods. *Journal of the American Society for Information Science and Technology*, *60*(3), 538–556.

Subotic, S., & Mukherjee, B. (2014). Short and amusing: The relationship between title characteristics, downloads, and citations in psychology articles. *Journal of Information Science*, *40*(1), 115–124.

Tabatabaei-Malazy, O., Ramezani, A., Atlasi, R., Larijani, B., & Abdollahi, M. (2016). Scientometric study of academic publications on antioxidative herbal medicines in type 2 diabetes mellitus. *Journal of Diabetes & Metabolic Disorders*, *15*(1), 48.

Taşkın, Z., & Doğan, G. (2013). Evaluation of scientific disciplines in Turkey: A citation analysis study. In *International symposium on information management in a changing world* (pp. 148–155). Springer, Berlin, Heidelberg.

Todorovsky, D. (2014). Follow-up study: On the working time budget of a university teacher. 45 years self-observation. *Scientometrics*, *101*(3), 2063–2070.

Weren, E. R., Kauer, A. U., Mizusaki, L., Moreira, V. P., de Oliveira, J. P. M., & Wives, L. K. (2014). Examining multiple features for author profiling. *Journal of Information and Data Management*, *5*(3), 266.

Weren, E. R., Moreira, V. P., & Oliveira, J. (2013). Using simple content features for the author profiling task. In *Notebook for PAN at cross-language evaluation forum*. Valencia, Spain.

Wolfram, D., & Zhao, Y. (2014). A comparison of journal similarity across six disciplines using citing discipline analysis. *Journal of Informetrics*, *8*(4), 840–853.

Zipf, G. K. (1932). *Selected studies of the principle of relative frequency in language*. Oxford: Harvard University Press.

Affiliations

Malik Muhammad Saad Missen[1] · Sajeeha Qureshi[1] · Nadeem Salamat[2] · Nadeem Akhtar[1] · Hina Asmat[1] · Mickaël Coustaty[3] · V. B. Surya Prasath[4,5,6,7]

1 Department of Computer Science & IT, The Islamia University, Bahawalpur 63100, Pakistan

2 Department of Mathematics, Khawaja Fareed University of Engineering and Technology, Rahim Yar Khan 64200, Pakistan

3 Laboratoire Informatique, Image et Interaction (L3i), Facultés des Sciences et Technologies, University of La Rochelle, La Rochelle, France

4 Division of Biomedical Informatics, Cincinnati Children's Hospital Medical Center, Cincinnati, OH 45229, USA

5 Department of Pediatrics, University of Cincinnati, Cincinnati, OH, USA

6 Department of Biomedical Informatics, College of Medicine, University of Cincinnati, Cincinnati, OH 45267, USA

7 Department of Electrical Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221, USA